VECTOR TRIPLE PRODUCT AND PRODUCT OF FOUR VECTORS

SYNOPSIS

- **Vector triple Product :**
The vector product of $\bar{a}\times\bar{b}$ and $\bar{c}$ is a vector triple Product of three vectors $\bar{a},\bar{b}$ and $\bar{c}$. It is denoted by $(\bar{a}\times\bar{b})\times\bar{c}$
- $(\bar{a}\times\bar{b})\times\bar{c}=(\bar{a}.\bar{c})\bar{b}-(\bar{b}.\bar{c})\bar{a}$. This is a vector in the plane of $\bar{a}$ and $\bar{b}$.
- $\bar{a}\times(\bar{b}\times\bar{c})=(\bar{a}.\bar{c})\bar{b}-(\bar{a}.\bar{b})\bar{c}$. This is a vector in the plane of $\bar{b},\bar{c}$
- $(\bar{a}\times\bar{b})\times\bar{c}=-\bar{c}\times(\bar{a}\times\bar{b})$
- $\bar{a},\bar{b},\bar{c}$ are non-zero vectors and $(\bar{a}\times\bar{b})\times\bar{c}=\bar{a}\times(\bar{b}\times\bar{c})\Leftrightarrow\bar{a}\,\&\,\bar{c}$ are collinear (Parallel) (or) $(\bar{a}\times\bar{c})\times\bar{b}=\bar{0}$
- Vector triple product is not associative. If $\bar{a},\bar{b},\bar{c}$ are non-zero, non-orthogonal vectors., then $(\bar{a}\times\bar{b})\times\bar{c}\neq\bar{a}\times(\bar{b}\times\bar{c})$.
- $\bar{a}\times(\bar{b}\times\bar{c})+\bar{b}\times(\bar{c}\times\bar{a})+\bar{c}\times(\bar{a}\times\bar{b})=\bar{0}$
$\Rightarrow\bar{a}\times(\bar{b}\times\bar{c}),\bar{b}\times(\bar{c}\times\bar{a}),\bar{c}\times(\bar{a}\times\bar{b})$ are coplanar
- $\bar{i}\times(\bar{j}\times\bar{k})+\bar{j}\times(\bar{k}\times\bar{i})+\bar{k}\times(\bar{i}\times\bar{j})=\bar{0}$
- $\bar{i}\times(\bar{a}\times\bar{i})+\bar{j}\times(\bar{a}\times\bar{i})+\bar{k}\times(\bar{a}\times\bar{k})=2\bar{a}$
where $\bar{a}$ is any vector
- $\left|\bar{a}\times(\bar{b}\times\bar{c})+\bar{b}\times(\bar{c}\times\bar{a})\right|=\left|(\bar{a}\times\bar{b})\times\bar{c}\right|$
- $[\bar{a}\times\bar{b}\ \ \bar{b}\times\bar{c}\ \ \bar{c}\times\bar{a}]=[\bar{a}\,\bar{b}\,\bar{c}]^2$
- **Scalar Product of Four Vectors :**
$(\bar{a}\times\bar{b}).(\bar{c}\times\bar{d})$ is a scalar product of four vectors. It is a dot product of the vectors $\bar{a}\times\bar{b}$ and $\bar{c}\times\bar{d}$.
- $(\bar{a}\times\bar{b}).(\bar{c}\times\bar{d})=(\bar{a}.\bar{c})(\bar{b}.\bar{d})-(\bar{a}.\bar{d})(\bar{b}.\bar{c})$
$$=\begin{vmatrix}\bar{a}.\bar{c} & \bar{a}.\bar{d}\\ \bar{b}.\bar{c} & \bar{b}.\bar{d}\end{vmatrix}$$
- $(\bar{a}\times\bar{c}).(\bar{b}\times\bar{d})=\begin{vmatrix}\bar{a}.\bar{b} & \bar{a}.\bar{d}\\ \bar{c}.\bar{b} & \bar{c}.\bar{d}\end{vmatrix}$
- $(\bar{a}\times\bar{b}).(\bar{a}\times\bar{b})=|\bar{a}\times\bar{b}|^2=|\bar{a}|^2|\bar{b}|^2-(\bar{a}.\bar{b})^2$
- **Vector Product of Four Vectors :**
$(\bar{a}\times\bar{b})\times(\bar{c}\times\bar{d})$ is a vector product of four vectors.
- $(\bar{a}\times\bar{b})\times(\bar{c}\times\bar{d})=[\bar{a}\,\bar{b}\,\bar{d}]\bar{c}-[\bar{a}\,\bar{b}\,\bar{c}]\bar{d}$
$=[\bar{c}\,\bar{d}\,\bar{a}]\bar{b}-[\bar{c}\,\bar{d}\,\bar{b}]\bar{a}$
- $[\bar{a}\,\bar{b}\,\bar{c}]\bar{d}=[\bar{b}\,\bar{c}\,\bar{d}]\bar{a}+[\bar{c}\,\bar{a}\,\bar{d}]\bar{b}+[\bar{a}\,\bar{b}\,\bar{d}]\bar{c}$
- $[\bar{a}\times\bar{b}\ \bar{b}\times\bar{c}\ \bar{c}\times\bar{a}]=[\bar{a}\,\bar{b}\,\bar{c}]^2$
- If $\bar{a},\bar{b},\bar{c}$ are non coplanar vectors, i.e., $[\bar{a}\,\bar{b}\,\bar{c}]\neq0$ then any vector $\bar{r}$ in space can be expressed as a linear combination of $\bar{a},\bar{b},\bar{c}$ i.e.,
$$\bar{r}=\frac{[\bar{r}\,\bar{b}\,\bar{c}]}{[\bar{a}\,\bar{b}\,\bar{c}]}\bar{a}+\frac{[\bar{r}\,\bar{c}\,\bar{a}]}{[\bar{a}\,\bar{b}\,\bar{c}]}\bar{b}+\frac{[\bar{r}\,\bar{a}\,\bar{b}]}{[\bar{a}\,\bar{b}\,\bar{c}]}\bar{c}$$
i.e., in the form $\bar{r}=x\bar{a}+y\bar{b}+z\bar{c}$
- If $\bar{a},\bar{b},\bar{c}$ and $\bar{d}$ are coplanar then $(\bar{a}\times\bar{b})\times(\bar{c}\times\bar{d})=\bar{0}$
- If $\bar{a},\bar{b},\bar{c}$ and $\bar{d}$ are parallel vectors (or) collinear vectors, then $(\bar{a}\times\bar{b})\times(\bar{c}\times\bar{d})=\bar{0}$

W.E-1: ***Let $\bar{a}=2\bar{i}+\bar{j}+\bar{k},\bar{b}=\bar{i}+2\bar{j}-\bar{k}$ and a unit vector $\bar{c}$ be coplanar. If $\bar{c}$ is perpendicular to $\bar{a}$, then $\bar{c}$ is equal to***

Sol: Required unit vector is $\pm\dfrac{\bar{a}\times(\bar{a}\times\bar{b})}{|\bar{a}\times(\bar{a}\times\bar{b})|}$

$\bar{a}\times(\bar{a}\times\bar{b})=(\bar{a}.\bar{b})\bar{a}-(\bar{a}.\bar{a})\bar{b}=-9\bar{j}+9\bar{k}$

$\therefore\bar{c}=\pm\dfrac{1}{\sqrt{2}}(-\bar{j}+\bar{k})$

W.E-2: *Let $\bar{a}=\bar{i}+\bar{j}$ and $\bar{b}=2\bar{i}-\bar{k}$ then point of intersection of the line $\bar{r}\times\bar{a}=\bar{b}\times\bar{a}$ and $\bar{r}\times\bar{b}=\bar{a}\times\bar{b}$ is*

Sol: We have $\bar{r}\times\bar{a}=\bar{b}\times\bar{a}\Rightarrow(\bar{r}-\bar{b})\times\bar{a}=\bar{0}$

$\Rightarrow\bar{r}-\bar{b}\parallel\bar{a}\Rightarrow\bar{r}-\bar{b}=\lambda\bar{a}$

$\Rightarrow\bar{r}=\bar{b}+\lambda\bar{a}$

Similarly, the equation of the line $\bar{r}\times\bar{b}=\bar{a}\times\bar{b}$ can be written as $\bar{r}=\bar{a}+\mu\bar{b}$

For the point of intersection of the above two lines, we have $\bar{a}+\mu\bar{b}=\bar{b}+\lambda\bar{a}\Rightarrow\lambda=\mu=1$

$\therefore\bar{r}=\bar{a}+\bar{b}=3\bar{i}+\bar{j}-\bar{k}$

W.E-3: $(\bar{b}\times\bar{c})\times(\bar{c}\times\bar{a})$ *is equal to*

Sol: $(\bar{b}\times\bar{c})\times(\bar{c}\times\bar{a})=\{(\bar{b}\times\bar{c}).\bar{a}\}\bar{c}-\{(\bar{b}\times\bar{c}).\bar{c}\}\bar{a}$

$=[\mathrm{a\ b\ c}]\bar{c}-[\mathrm{b\ c\ c}]\bar{a}=[\mathrm{a\ b\ c}]\bar{c}$

LEVEL - I (C.W)

VECTOR TRIPLE PRODUCT

1. If $\bar{a}=\bar{i}+\bar{j}+\bar{k},\bar{b}=\bar{i}-\bar{j}+\bar{k},\bar{c}=2\bar{i}+3\bar{j}-\bar{k}$, then $(\bar{a}\times\bar{b})\times\bar{c}=$
   1) $2\bar{i}-6\bar{j}+2\bar{k}$ 2) $6\bar{i}-2\bar{j}+6\bar{k}$
   3) $-6\bar{i}+2\bar{j}-6\bar{k}$ 4) $6\bar{i}+2\bar{j}+6\bar{k}$
2. If $\bar{a}=\bar{i}+\bar{j}+\bar{k},\bar{b}=\bar{i}+\bar{j},\bar{c}=\bar{i}$ and $(\bar{a}\times\bar{b})\times\bar{c}=\lambda\bar{a}+\mu\bar{b}$, then $\lambda+\mu=$
   1) 1 2) 0 3) -1 4) 2
3. $\bar{a}=2\bar{i}+3\bar{j}-4\bar{k},\bar{b}=\bar{i}+\bar{j}+\bar{k}$, $\bar{c}=4\bar{i}+2\bar{j}+3\bar{k}$ then $|\bar{a}\times(\bar{b}\times\bar{c})|=$ **(EAM-2000)**
   1) $\sqrt{10}$ 2) 1 3) 2 4) $\sqrt{5}$
4. $\bar{a}\times(\bar{b}\times\bar{c})+\bar{b}\times(\bar{c}\times\bar{a})+\bar{c}\times(\bar{a}\times\bar{b})=$
   1) 0 2) $\bar{0}$ 3) 1 4) $(\bar{a}\times\bar{b}).\bar{c}$
5. $(\bar{a}\times\bar{b})\times\bar{c}=\bar{a}\times(\bar{b}\times\bar{c})$ if and only if
   1) $(\bar{a}\times\bar{c})\times\bar{b}=\bar{0}$ 2) $\bar{a}\times(\bar{c}\times\bar{b})=\bar{0}$
   3) $\bar{c}\times(\bar{b}\times\bar{a})=\bar{0}$ 4) $[\bar{a}\,\bar{b}\,\bar{c}]=1$
6. The vector $(\bar{a}\times\bar{b})\times\bar{c}$ is perpendicular to
   1) $\bar{c}$ 2) $\bar{a}\times\bar{b}$ 3) both 1 and 2 4) $\bar{b},\bar{c}$
7. $\bar{i}\times(\bar{a}\times\bar{i})+\bar{j}\times(\bar{a}\times\bar{j})+\bar{k}\times(\bar{a}\times\bar{k})=$
   1) $3\bar{a}$ 2) $2\bar{a}$ 3) $\bar{a}$ 4) $\bar{0}$

SCALAR PRODUCT OF FOUR VECTORS:

8. $\bar{a}=2\bar{i}+3\bar{j}-\bar{k},\bar{b}=-\bar{i}+2\bar{j}-4\bar{k}$, $\bar{c}=\bar{i}+\bar{j}+\bar{k}$, $\bar{d}=\bar{i}-\bar{j}-\bar{k}$ then $(\bar{a}\times\bar{b}).(\bar{c}\times\bar{d})=$ ___
   1) 4 2) 24 3) 36 4) 4
9. If $(\bar{a}\times\bar{b})(\bar{c}\times\bar{b})+(\bar{a}.\bar{b})(\bar{b}.\bar{c})=\lambda(\bar{a}.\bar{c})$ then $\lambda=$
   1) $|\bar{a}|^2$ 2) $|\bar{b}|^2$ 3) $|\bar{c}|^2$ 4) 0
10. $(\bar{a}\times\bar{i})(\bar{b}\times\bar{i})+(\bar{a}\times\bar{j})(\bar{b}\times\bar{j})+(\bar{a}\times\bar{k})(\bar{b}\times\bar{k})=$
   1) $\bar{a}.\bar{b}$ 2) $3(\bar{a}.\bar{b})$ 3) 0 4) $2(\bar{a}.\bar{b})$
11. If $\bar{a}$ is parallel to $\bar{b}\times\bar{c}$, then $(\bar{a}\times\bar{b})(\bar{a}\times\bar{c})=$
   1) $|\bar{a}|^2(\bar{b}.\bar{c})$ 2) $|\bar{b}|^2(\bar{a}.\bar{c})$
   3) $|\bar{c}|^2(\bar{a}.\bar{b})$ 4) 0
12. If $\bar{a},\bar{b}$ are two unit vectors such that $|\bar{a}\times\bar{b}|=2$ then the value of $[\bar{a}\ \ \bar{b}\ \ \bar{a}\times\bar{b}]$ is
   1) 1 2) 2 3) 4 4) 0

VECTOR PRODUCT OF FOUR VECTORS:

13. If $\bar{a}=2\bar{i}+\bar{j}+3\bar{k},\bar{b}=3\bar{i}+2\bar{j}+\bar{k}$, $\bar{c}=\bar{i}-\bar{j}-4\bar{k},\bar{d}=\bar{i}+2\bar{j}-\bar{k}$ then $(\bar{a}\times\bar{b})\times(\bar{c}\times\bar{d})=$
   1) $24(\bar{i}+\bar{j}-2\bar{k})$ 2) $24(\bar{i}-\bar{j}-\bar{k})$
   3) $12(2\bar{i}+\bar{j}-3\bar{k})$ 4) $12(\bar{i}-2\bar{j}+3\bar{k})$
14. If four vectors $\bar{a},\bar{b},\bar{c},\bar{d}$ are coplanar, then $(\bar{a}\times\bar{b})\times(\bar{c}\times\bar{d})=$
   1) $[\bar{a}\,\bar{b}\,\bar{c}]\bar{d}$ 1) $[\bar{b}\,\bar{c}\,\bar{d}]\bar{a}$
   3) $[\bar{c}\ \bar{d}\ \bar{a}]\bar{b}$ 4) Null vector

15. **If** $\left(\bar{b}\times\bar{c}\right)\times\left(\bar{c}\times\bar{a}\right)=3\bar{c}$ **then** $\left[\bar{b}\times\bar{c}\quad\bar{c}\times\bar{a}\quad\bar{a}\times\bar{b}\right]=$

1) 2 2) 7 3) 9 4) 11

LEVEL-I(C.W)-KEY

01) 2	02) 2	03) 4	04) 2	05) 1	06) 3
07) 2	08) 4	09) 2	10) 4	11) 1	12) 3
13) 1	14) 4	15) 3			

LEVEL-I(C.W)-HINTS

1. $\left(\bar{a}.\bar{c}\right)\bar{b}-\left(\bar{c}.\bar{b}\right)\bar{a}$

2. $\left(\bar{a}\times\bar{c}\right)\times\bar{b}=\lambda\bar{a}+\mu\bar{b}$

$\Rightarrow\left(\bar{a}.\bar{c}\right)\bar{b}-\left(\bar{b}.\bar{c}\right)\bar{a}=\lambda\bar{a}+\mu\bar{b}$

$\Rightarrow\lambda=-\left(\bar{b}.\bar{c}\right),\ \mu=\left(\bar{c}.\bar{a}\right)$

3. $\left|\left(\bar{a}.\bar{c}\right)\bar{b}-\left(\bar{a}.\bar{b}\right)\bar{c}\right|$

and $\left|a_1\bar{i}+a_2\bar{j}+a_3\bar{k}\right|=\sqrt{a_1^2+a_2^2+a_3^2}$

4. $\sum\left(\bar{a}\times\left(\bar{b}\times\bar{c}\right)\right)=\sum\left(\left(\bar{a}.\bar{c}\right)\bar{b}-\left(\bar{a}.\bar{b}\right)\bar{c}\right)=\bar{0}$

5. $\left(\bar{a}\times\bar{b}\right)\times\bar{c}=\bar{a}\times\left(\bar{b}\times\bar{c}\right)$

$\Rightarrow\left(\bar{c}.\bar{a}\right)\bar{b}-\left(\bar{c}.\bar{b}\right)\bar{a}=\left(\bar{a}.\bar{c}\right)\bar{b}-\left(\bar{a}.\bar{b}\right)\bar{c}$

$\Rightarrow\left(\bar{a}.\bar{b}\right)\bar{c}-\left(\bar{c}.\bar{b}\right)\bar{a}=0\quad\Rightarrow\left(\bar{a}\times\bar{c}\right)\times\bar{b}=0$

6. Cross product of any two vectors is perpendicular to both the vectors

7. $\sum\bar{i}\times\left(\bar{a}\times\bar{i}\right)=\sum\left(\left(\bar{i}.\bar{i}\right)\bar{a}-\left(\bar{i}.\bar{a}\right)\bar{i}\right)$

$=\sum\left(\bar{a}-\left(\bar{a}.\bar{i}\right)\bar{i}\right)=3\bar{a}-\bar{a}=2\bar{a}$

8. $\begin{vmatrix}\bar{a}.\bar{c} & \bar{a}.\bar{d}\\ \bar{b}.\bar{c} & \bar{b}.\bar{d}\end{vmatrix}$

9. $\left(\bar{a}.\bar{c}\right)\left(\bar{b}.\bar{b}\right)-\left(\bar{a}.\bar{b}\right)\left(\bar{b}.\bar{c}\right)+\left(\bar{a}.\bar{b}\right)\left(\bar{b}.\bar{c}\right)=\lambda\left(\bar{a}.\bar{c}\right)$

$\Rightarrow\left(\bar{a}.\bar{c}\right)\left|\bar{b}\right|^2=\lambda\left(\bar{a}.\bar{c}\right)\qquad\therefore\lambda=\left|\bar{b}\right|^2$

10. Use scalar product of four vectors formula

11. $\begin{vmatrix}\bar{a}.\bar{a} & \bar{a}.\bar{c}\\ \bar{b}.\bar{a} & \bar{b}.\bar{c}\end{vmatrix}=\left|\bar{a}\right|^2\left(\bar{b}.\bar{c}\right)-\left(\bar{b}.\bar{a}\right)\left(\bar{a}.\bar{c}\right)$

$=\left|\bar{a}\right|^2\left(\bar{b}.\bar{c}\right)\quad\left(\because\bar{b}.\bar{a}=\bar{c}.\bar{a}=0\right)$

12. $\left[\bar{a}\ \bar{b}\ \bar{a}\times\bar{b}\right]=\left(\bar{a}\times\bar{b}\right).\left(\bar{a}\times\bar{b}\right)=\left|\bar{a}\times\bar{b}\right|^2$

13. Find $\left[\bar{a}\ \bar{b}\ \bar{d}\right]\bar{c}-\left[\bar{a}\ \bar{b}\ \bar{c}\right]\bar{d}$

14. $\left[\bar{a}\ \bar{b}\ \bar{d}\right]\bar{c}-\left[\bar{a}\ \bar{b}\ \bar{c}\right]\bar{d}=\bar{0}-\bar{0}=\bar{0}$

15. $\left[\bar{b}\ \bar{c}\ \bar{a}\right]\bar{c}=3\bar{c}\qquad\Rightarrow\left[\bar{a}\ \bar{b}\ \bar{c}\right]=3$

Required value $\left[\bar{a}\ \bar{b}\ \bar{c}\right]^2=9$

LEVEL - I (H.W)

VECTOR TRIPLE PRODUCT

1. **If** $\bar{a}=\bar{i}+\bar{j}-\bar{k},\ \bar{b}=\bar{i}-\bar{j}+\bar{k},$ $\bar{c}=\bar{i}-\bar{j}-\bar{k}$ **then** $\bar{a}\times\left(\bar{b}\times\bar{c}\right)=$

1) $\bar{i}-\bar{j}+\bar{k}$ 2) $2\bar{i}-2\bar{j}$
3) $3\bar{i}-\bar{j}+\bar{k}$ 4) $2\bar{i}+2\bar{j}-\bar{k}$

2. **If** $\bar{a}=\bar{i}-2\bar{j}-3\bar{k},\bar{b}=2\bar{i}+\bar{j}-\bar{k}$ **and** $\bar{c}=\bar{i}+3\bar{j}-2\bar{k}$, **and** $\bar{a}\times\left(\bar{b}\times\bar{c}\right)=p\bar{i}+q\bar{j}+r\bar{k}$, **then** $p+q+r=$

a) -4 b) 4 c) 2 d) -2

3. **If** $\bar{p}=(2,-10,2),\ \bar{q}=(3,1,2)$ **and** $\bar{r}=(2,1,3)$, **then** $\left|\bar{p}\times(\bar{q}\times\bar{r})\right|=$

a) 2 b) 4 c) 0 d) 3

4. $\bar{i}\times(\bar{j}\times\bar{k})+\bar{j}\times(\bar{k}\times\bar{i})+\bar{k}\times(\bar{i}\times\bar{j})=$

1) $\bar{i}$ 2) $\bar{j}$ 3) $\bar{k}$ 4) Null vector

5. **If** $\left(\bar{a}\times\bar{b}\right)\times\bar{c}=\bar{a}\times\left(\bar{b}\times\bar{c}\right)$ **where** $\bar{a}$ **,** $\bar{b}$ **and** $\bar{c}$ **are any three vectors such that** $\bar{a}.\bar{b}\neq0$, $\bar{b}.\bar{c}\neq0$ **then** $\bar{a}$ **and** $\bar{c}$ **are**

1) Perpendicular 2) Parallel
3) Inclined at an angle of $\frac{\pi}{3}$ between them
4) Inclined at an angle of $\frac{\pi}{6}$ between tehm

6. **The vectors** $\bar{a}\times\left(\bar{b}\times\bar{c}\right)$ **is**

1) Coplanar with $\bar{b}$ and $\bar{c}$
2) Coplanar with $\bar{a}$ and $\bar{b}$ parallel to $\bar{c}$
3) Coplanar with $\bar{b}$ and $\bar{c}$, orthogonal to $\bar{a}$
4) Coplanar with $\bar{a}$ and $\bar{b}$, orthogonal to $\bar{c}$

7. $\left(\bar{a}\times\bar{i}\right)\times\bar{i}+\left(\bar{a}\times\bar{j}\right)\times\bar{j}+\left(\bar{a}\times\bar{k}\right)\times\bar{k}=$ ___

1) $2\bar{a}$ 2) $-2\bar{a}$ 3) $\bar{a}$ 4) $-\bar{a}$

SCALAR PRODUCT OF FOUR VECTORS

8. If $\bar{a}=\bar{i}+\bar{j}+\bar{k}, \bar{b}=\bar{i}+\bar{j}-\bar{k}$, $\bar{c}=\bar{i}-\bar{j}+\bar{k}$, $\bar{d}=\bar{i}-\bar{j}-\bar{k}$ then value of $(\bar{a}\times\bar{b}).(\bar{c}\times\bar{d})$ is

1) 1 2) 0 3) -2 4) -1

9. $(\bar{a}\times\bar{b}).(\bar{c}\times\bar{d})=(\bar{a}.\bar{c})(\bar{b}.\bar{d})+K(\bar{a}.\bar{d})(\bar{b}.\bar{c})$ then the value of K is

1) 1 2) 0 3) -2 4) -1

10. $(\bar{b}\times\bar{c}).(\bar{a}\times\bar{d})+(\bar{c}\times\bar{a}).(\bar{b}\times\bar{d})+(\bar{a}\times\bar{b}).(\bar{c}\times\bar{d})=$

1) 0 2) 1 3) 2 4) -1

11. $(\bar{a}\times\bar{b}).(\bar{c}\times\bar{d})+(\bar{b}.\bar{c})(\bar{a}.\bar{d})=$

1) 0 2) 1 3) $\bar{a}.\bar{c}$ 4) $(\bar{a}.\bar{c})(\bar{b}.\bar{d})$

12. If $\bar{a}$, $\bar{b}$ lie in a plane normal to the plane containing $\bar{c}$ and $\bar{d}$ then $|(\bar{a}\times\bar{b}).(\bar{c}\times\bar{d})|=$

1) 4 2) 1 3) 0 4) 3

VECTOR PRODUCT OF FOUR VECTORS

13. $\bar{a}=2\bar{i}+3\bar{j}-\bar{k}, \bar{b}=-\bar{i}+2\bar{j}-4\bar{k}, \bar{c}=\bar{i}+\bar{j}+\bar{k}$, $\bar{d}=3\bar{i}-4\bar{j}+2\bar{k}$. The value of $(\bar{a}\times\bar{b})\times(\bar{c}\times\bar{d})=$

1) $-70\bar{i}-28\bar{j}-64\bar{k}$ 2) $7\bar{i}+8\bar{j}$

3) $6\bar{j}-2\bar{k}$ 4) $\bar{i}+\bar{j}-4\bar{k}$

14. $(\bar{a}\times\bar{b})\times(\bar{c}\times\bar{d})=l\bar{c}+m\bar{d}$ then m is

1) $[\bar{a}\ \bar{b}\ \bar{c}\ \bar{d}]$ 2) $[\bar{c}\ \bar{b}\ \bar{d}]$

3) $[\bar{b}\ \bar{c}\ \bar{d}]$ 4) $-[\bar{a}\ \bar{b}\ \bar{c}]$

15. If $\alpha\bar{a}+\beta\bar{b}+\gamma\bar{c}=\bar{0}$ then $(\bar{a}\times\bar{b})\times\{(\bar{b}\times\bar{c})\times(\bar{c}\times\bar{a})\}=$

1) $\bar{0}$

2) A vector perpendicular to the plane of $\bar{a},\bar{b},\bar{c}$

3) A scalar quantity 4) $2[\bar{a}\ \bar{b}\ \bar{c}]$

LEVEL-I(H.W)-KEY

01) 2	02) 1	03) 3	04) 4	05) 2	06) 3
07) 2	08) 2	09) 4	10) 1	11) 4	12) 3
13) 1	14) 4	15) 1			

LEVEL-I(H.W)-HINTS

1. $(\bar{a}.\bar{c})\bar{b}-(\bar{a}.\bar{b})\bar{c}$

2. $(\bar{a}.\bar{c})\bar{b}-(\bar{a}.\bar{b})\bar{c}$

3. $|(\bar{p}.\bar{r})\bar{q}-(\bar{p}.\bar{q})\bar{r}|$

4. $\bar{a}\times(\bar{b}\times\bar{c})=(\bar{a}.\bar{c})\bar{b}-(\bar{a}.\bar{b})\bar{c}$

5. $(\bar{c}.\bar{a})\bar{b}-(\bar{c}.\bar{b})\bar{a}=(\bar{a}.\bar{c})\bar{b}-(\bar{a}.\bar{b})\bar{c}$

$\Rightarrow(\bar{c}.\bar{b})\bar{a}=(\bar{a}.\bar{b})\bar{c}$ $\Rightarrow\bar{a},\bar{c}$ are parallel

6. Cross product of any two vectors is perpendicular to both the vectors and

$(\bar{a}\times\bar{b})\times\bar{c}=(\bar{c}.\bar{a})\bar{b}-(\bar{c}.\bar{b})\bar{a}$

7. $\sum(\bar{a}\times\bar{i})\times\bar{i}=-\sum\bar{i}\times(\bar{a}\times\bar{i})$

$=-\sum((\bar{i}.\bar{i})\bar{a}-(\bar{i}.\bar{a})\bar{i})$

$=-\sum(\bar{a}-(\bar{i}.\bar{a})\bar{i})$ $=-3\bar{a}+\bar{a}=-2\bar{a}$

8. $\begin{vmatrix}\bar{a}.\bar{c} & \bar{a}.\bar{d}\\ \bar{b}.\bar{c} & \bar{b}.\bar{d}\end{vmatrix}$

9. $(\bar{a}.\bar{c})(\bar{b}.\bar{d})-(\bar{a}.\bar{d})(\bar{b}.\bar{c})=(\bar{a}.\bar{c})(\bar{b}.\bar{d})+K(\bar{a}.\bar{d})(\bar{b}.\bar{c})$

$\Rightarrow K=-1$

10. Use scalar product of four vectors formula

11. $\begin{vmatrix}\bar{a}.\bar{c} & \bar{a}.\bar{d}\\ \bar{b}.\bar{c} & \bar{b}.\bar{d}\end{vmatrix}+(\bar{b}.\bar{c})(\bar{a}.\bar{d})$

12. $(\bar{a}\times\bar{b})\perp(\bar{c}\times\bar{d})\Rightarrow(\bar{a}\times\bar{b}).(\bar{c}\times\bar{d})=0$

13. $\bar{a}\times\bar{b}=-10\bar{i}+9\bar{j}+7\bar{k}$

$\bar{c}\times\bar{d}=6\bar{i}+\bar{j}-7\bar{k}$

$(\bar{a}\times\bar{b})\times(\bar{c}\times\bar{d})=-70\bar{i}-28\bar{j}-64\bar{k}$

14. $\Rightarrow[\bar{a}\ \bar{b}\ \bar{d}]\bar{c}-[\bar{a}\ \bar{b}\ \bar{c}]\bar{d}=l\bar{c}+m\bar{d}$

$\therefore m=-[\bar{a}\ \bar{b}\ \bar{c}]$

15. The vectors $\bar{a}$, $\bar{b}$, $\bar{c}$ are coplanar

$\Rightarrow \bar{b}\times\bar{c}$ and $\bar{c}\times\bar{a}$ are parallel

$\Rightarrow (\bar{b}\times\bar{c})\times(\bar{c}\times\bar{a}) = \bar{0} \quad \therefore (\bar{a}\times\bar{b})\times\bar{0} = \bar{0}$

LEVEL - II (C.W)

VECTOR TRIPLE PRODUCT

1. $\bar{a}\times(\bar{b}\times\bar{c})$ **is parallel to** $\bar{b}$**, then**

1) $\bar{a}\parallel\bar{c}$ 2) $\bar{b}\parallel\bar{c}$ 3) $\bar{a}\parallel\bar{b}$

4) $\bar{a},\bar{b}$ & $\bar{c}$ are parallel to each other

2. $\bar{i}\times\left[(\bar{a}\times\bar{b})\times\bar{i}\right]+\bar{j}\times\left[(\bar{a}\times\bar{b})\times\bar{j}\right]+\bar{k}\times\left[(\bar{a}\times\bar{b})\times\bar{k}\right]=$

1) $\bar{0}$ 2) $(\bar{a}\cdot\bar{b})\bar{b}$ 3) $\bar{b}$ 4) $2(\bar{a}\times\bar{b})$

3. $\bar{a},\bar{b},\bar{c}$ **are three unit vectors such that**

$\bar{a}\times(\bar{b}\times\bar{c})=\frac{1}{2}\bar{b}$**, then** $(\bar{a},\bar{b})=$, $(\bar{a},\bar{c})=$

($\bar{b},\bar{c}$ are non-collinear)

1) $90^0,60^0$ 2) $60^0,90^0$ 3) $30^0,60^0$ 4) $45^0,30^0$

4. If $\bar{a},\bar{b},\bar{c}$ **are non-coplanar unit vectors such that** $\bar{a}\times(\bar{b}\times\bar{c})=\frac{\bar{b}+\bar{c}}{\sqrt{2}}$**, then the angle between** $\bar{a}$ **and** $\bar{b}$ **is**

1) $\frac{3\pi}{4}$ 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{2}$ 4) π

5. $\bar{a}=2\bar{i}+\bar{j}-2\bar{k}$ **and** $\bar{b}=\bar{i}+\bar{j}$ **if** $\bar{c}$ **is a vector such that** $\bar{a}\cdot\bar{c}=|\bar{c}|,|\bar{c}-\bar{a}|=2\sqrt{2}$ **and the angle between** $\bar{a}\times\bar{b}$ **and** $\bar{c}$ **is** 30^0**, then** $|(\bar{a}\times\bar{b})\times\bar{c}|=$

1) 2/3 2) 3/2 3) 2 4) 3

6. If $\bar{a}$ *and* $\bar{b}$ **are unit vectors then the vectors** $(\bar{a}+\bar{b})\times(\bar{a}\times\bar{b})$ **is parallel to the vector**

1) $\bar{a}-\bar{b}$ 2) $\bar{a}+\bar{b}$ 3) $2\bar{a}-\bar{b}$ 4) $2\bar{a}+\bar{b}$

7. If $\bar{a}=2\bar{i}+\bar{j}+\bar{k}$, $\bar{b}=\bar{i}+2\bar{j}+2\bar{k}$, $\bar{c}=\bar{i}+\bar{j}+2\bar{k}$ **and** $\bar{a}\times(\bar{b}\times\bar{c})=(1+\alpha)\bar{i}+\beta(1+\alpha)\bar{j}+\gamma(1+\alpha)(1+\beta)\bar{k}$ **then**

1) $\alpha=-2,\beta=-4,\gamma=\frac{-2}{3}$ 2) $\alpha=2,\beta=-4,\gamma=\frac{2}{3}$

3) $\alpha=-2,\beta=4,\gamma=\frac{2}{3}$ 4) $\alpha=2,\beta=4,\gamma=\frac{-2}{3}$

8. If $\bar{a}$ **is a unit vector then** $\bar{a}\times\{\bar{a}\times(\bar{a}\times\bar{b})\}=$

1) $\bar{a}\times\bar{b}$ 2) $\bar{b}\times\bar{a}$ 3) $(\bar{a}\times\bar{b})$ 4) $2(\bar{b}\times\bar{a})$

9. If $\bar{a},\bar{b},\bar{c}$ **are three vectors of magnitude** $\sqrt{3},1,2$ **such that** $\bar{a}\times(\bar{a}\times\bar{c})+3\bar{b}=\bar{0}$ **and** θ **is the angle between** $\bar{a}$ **and** $\bar{c}$ **then** $\cos^2\theta=$

1) $\frac{3}{4}$ 2) $\frac{1}{2}$ 3) $\frac{1}{4}$ 4) $\frac{1}{5}$

10. If $\bar{a}=\bar{i}+\bar{j}-2\bar{k}$ **then** $\sum\left[(\bar{a}\times\bar{i})\times\bar{j}\right]^2=$

1) 6 2) 36 3) $\sqrt{6}$ 4) $6\sqrt{6}$

11. $(\bar{a}\times\bar{b})^2+(\bar{a}.\bar{b})^2=144$ and $|\bar{a}|=4$, $|\bar{b}|=$___

1) 16 2) 8 3) 3 4) 12

LEVEL-II-(C.W)-KEY

01) 2	02) 4	03) 1	04) 1	05) 2	06) 1
07) 1	08) 2	09) 1	10) 1	11) 3	

LEVEL-II-(C.W)-HINTS

1. $(\bar{a}\times(\bar{b}\times\bar{c}))\times\bar{b}=\bar{0}$

$\Rightarrow(\bar{a}.\bar{b})(\bar{b}\times\bar{c})=\bar{0} \quad \Rightarrow \bar{b}\parallel\bar{c}$

2. $\sum(\bar{i}\times(\bar{p}\times i))=2\bar{p}$

3. $\bar{a}\times(\bar{b}\times\bar{c})=\frac{1}{2}\bar{b}$

$\Rightarrow(\bar{a}.\bar{c})\bar{b}-(\bar{a}.\bar{b})\bar{c}=\frac{1}{2}\bar{b}-0(\bar{c})$

$\Rightarrow(\bar{a}.\bar{c})=\frac{1}{2},\ \bar{a}.\bar{b}=0 \quad \Rightarrow|\bar{a}||\bar{c}|\cos(\bar{a},\bar{c})=\frac{1}{2}$

$\Rightarrow\cos(\bar{a},\bar{c})=\frac{1}{2} \quad \therefore(\bar{a},\bar{c})=\frac{\pi}{3}$

$\bar{a}.\bar{b}=0\Rightarrow(\bar{a},\bar{b})=\frac{\pi}{2}$

4. $(\bar{a}.\bar{c})\bar{b}-(\bar{a}.\bar{b})\bar{c}=\frac{1}{\sqrt{2}}\bar{b}+\frac{1}{\sqrt{2}}\bar{c}$

$\Rightarrow \bar{a}.\bar{c}=\frac{1}{\sqrt{2}},\ \bar{a}.\bar{b}=\frac{-1}{\sqrt{2}}$

$\Rightarrow (\bar{a},\bar{c})=\frac{\pi}{4},\quad (\bar{a},\bar{b})=\frac{3\pi}{4}$

5. $|\bar{a}|=3,\ |\bar{a}\times\bar{b}|=|2\bar{i}+2\bar{j}+\bar{k}|=3$

$|\bar{c}-\bar{a}|=2\sqrt{2}$

$\Rightarrow |\bar{c}-\bar{a}|^2=8 \qquad \Rightarrow |\bar{c}|^2+|\bar{a}|^2-2(\bar{a}.\bar{c})=8$

$\Rightarrow c^2+9-2c=8 \qquad \because \bar{a}.\bar{c}=|\bar{c}|$

$\Rightarrow c^2-2c+1=0 \Rightarrow (c-1)^2=0$

$\therefore c=1 \qquad \therefore |\bar{c}|=1$

$|(\bar{a}\times\bar{b})\times\bar{c}|=|\bar{a}\times\bar{b}||\bar{c}|\sin(30)=3\times1\times\frac{1}{2}=\frac{3}{2}$

6. $(\bar{a}+\bar{b})\times(\bar{a}\times\bar{b})$

$=((\bar{a}+\bar{b}).\bar{b})\bar{a}-((\bar{a}+\bar{b}).\bar{a})\bar{b}$

$=(\bar{a}.\bar{b})\bar{a}+(\bar{b}.\bar{b}).\bar{a}-(\bar{a}.\bar{a})\bar{b}-(\bar{b}.\bar{a})\bar{b}$

$=(\bar{a}.\bar{b})\bar{a}+\bar{a}-\bar{b}-(\bar{a}.\bar{b})\bar{b}$

$=\{(\bar{a}.\bar{b})+1\}(\bar{a}-\bar{b})$

7. $\bar{a}\times(\bar{b}\times\bar{c})=-\bar{i}+4\bar{j}-2\bar{k}$

$1+\alpha=-1,\ \beta(1+\alpha)=4, \gamma(1+\alpha)(1+\beta)=-2$

8. $\Rightarrow \bar{a}\times\{(\bar{a}.\bar{b})\bar{a}-(\bar{a}.\bar{a})\bar{b}\}$

$=(\bar{a}.\bar{b})(\bar{a}\times\bar{a})-(\bar{a}.\bar{a})(\bar{a}\times\bar{b}) \ =\bar{b}\times\bar{a}$

9. $(\bar{a}.\bar{c})\bar{a}-(\bar{a}.\bar{a})\bar{c}+3\bar{b}=\bar{0}$

$\Rightarrow 2\sqrt{3}\cos\theta\bar{a}-3\bar{c}+3\bar{b}=\bar{0}$

$\Rightarrow 2\cos\theta\bar{a}-\sqrt{3}\bar{c}+\sqrt{3}\bar{b}=\bar{0}$

$\Rightarrow |2\cos\theta\bar{a}-\sqrt{3}\bar{c}|^2=(-\sqrt{3}\bar{b})^2$

$\Rightarrow 12\cos^2\theta+9-24\cos^2\theta=0 \ \Rightarrow \cos^2\theta=\frac{3}{4}$

10. $\Rightarrow |\bar{a}|^2=1+1+4=6$

11. $|\bar{a}\times\bar{b}|^2=|\bar{a}|^2|\bar{b}|^2-(\bar{a}.\bar{b})^2$

LEVEL - II (H.W)

VECTOR TRIPLE PRODUCT

1. Which of the following is a true statement.

1) $(\bar{a}\times\bar{b})\times\bar{c}$ is coplanar with $\bar{c}$

2) $(\bar{a}\times\bar{b})\times\bar{c}$ is perpendicular to $\bar{a}$

3) $(\bar{a}\times\bar{b})\times\bar{c}$ is perpendicular to $\bar{b}$

4) $(\bar{a}\times\bar{b})\times\bar{c}$ is perpendicular to $\bar{c}$

2. If $\bar{a}=\bar{i}+\bar{j}+\bar{k},\bar{b}=\bar{i}-\bar{j}+\bar{k}$ then

$\bar{i}\times[(\bar{a}\times\bar{b})\times\bar{i}]+\bar{j}\times[(\bar{a}\times\bar{b})\times\bar{j}]+\bar{k}\times[(\bar{a}\times\bar{b})\times\bar{k}]=$

1) $2(\bar{i}+\bar{j})$ 2) $2(\bar{i}+\bar{k})$ 3) $4(\bar{i}-\bar{j})$ 4) $2(\bar{j}-\bar{k})$

3. Let $\bar{a},\bar{b},\bar{c}$ be three vectors having magnitude 1,1 and 2 respectively. If $\bar{a}\times(\bar{a}\times\bar{c})+\bar{b}=\bar{0}$, then the acute angle between $\bar{a}$ and $\bar{c}$ is

1) 30^0 2) 60^0 3) 45^0 4) 75^0

4. $\bar{b},\bar{c}$ are unit vectors and $|\bar{a}|=7$.

$\bar{a}\times(\bar{b}\times\bar{c})+\bar{b}\times(\bar{c}\times\bar{a})=\frac{1}{2}\bar{a}$ then the angle between $\bar{a}$ and $\bar{c}$ is

1) $\frac{\pi}{3}$ 2) $\frac{\pi}{6}$ 3) $\frac{\pi}{2}$ 4) $\frac{\pi}{4}$

5. If $\bar{a},\bar{b},\bar{c}$ are unit vectors and $\bar{b},\bar{c}$ are non collinear vectors satisfying.

$(\bar{a},\bar{b})=\alpha,(\bar{a},\bar{c})=\beta$ and $\bar{a}\times(\bar{b}\times\bar{c})=\frac{\bar{b}+\bar{c}}{2}$

then $\cos(\alpha+\beta)=$

1) 0 2) 1 3) -1 4) $\frac{1}{2}$

6. If $\bar{a},\bar{b},\bar{c}$ are any three vectors such that $(\bar{a}+\bar{b}).\bar{c}=(\bar{a}-\bar{b}).\bar{c}=0$ then $(\bar{a}\times\bar{b})\times\bar{c}=$

1) $\bar{0}$ 2) $\bar{a}$ 3) $\bar{b}$ 4) $\bar{c}$

7. If $\left(\bar{a}\times\bar{b}\right)\times\bar{c}=-5\bar{a}+4\bar{b}$ and $\bar{a}.\bar{b}=3$ then $\bar{a}\times\left(\bar{b}\times\bar{c}\right)=$

1) $5\bar{b}-3\bar{c}$ 2) $3\bar{c}-4\bar{b}$ 3) $3\bar{b}-5\bar{c}$ 4) $4\bar{b}-3\bar{c}$

8. $\sum\left(\left(\bar{a}\times\bar{i}\right)\times\bar{j}\right)^2=$

1) $\bar{a}^2$ 2) $2\bar{a}^2$ 3) $3\bar{a}^2$ 4) $4\bar{a}^2$

9. $\bar{a}\times\left\{\bar{a}\times\left(\bar{a}\times\bar{b}\right)\right\}=$

1) $\left(\bar{a}.\bar{b}\right)\left(\bar{a}\times\bar{b}\right)$ 2) $\left(\bar{a}.\bar{a}\right)\left(\bar{a}\times\bar{b}\right)$

3) $\left(\bar{a}.\bar{a}\right)\left(\bar{b}\times\bar{a}\right)$ 4) $\left(\bar{b}.\bar{b}\right)\left(\bar{a}\times\bar{b}\right)$

10. If $\bar{a}\times\left(\bar{a}\times\bar{b}\right)=\bar{b}\times\left(\bar{b}\times\bar{c}\right)$ and $\left(\bar{a}.\bar{b}\right)\neq 0$, then $\left[\bar{a}\ \bar{b}\ \bar{c}\right]=$

1) 1 2) 2 3) 3 4) 0

11. $\left[\bar{b}\ \bar{c}\ \bar{b}\times\bar{c}\right]+\left(\bar{b}.\bar{c}\right)^2=$___

1) $\left(\bar{b}+\bar{c}\right)^2$ 2) $\left|\bar{b}\right|^2+\left|\bar{c}\right|^2$ 3) $\left|\bar{b}\right|+\left|\bar{c}\right|$ 4) $\left|\bar{b}\right|^2\left|\bar{c}\right|^2$

12. If $\bar{a}=\bar{i}+\bar{j}+\bar{k}$, $\bar{b}=\bar{i}+\bar{j}-\bar{k}$, $\bar{c}=\bar{i}-\bar{j}+\bar{k}$, $\bar{d}=\bar{i}-\bar{j}-\bar{k}$ then $\left|\left(\bar{a}\times\bar{b}\right)\times\left(\bar{c}\times\bar{d}\right)\right|=$

1) 8 2) 1 3) 2 4) 3

13. $\left[\bar{a}\times\bar{b}\ \ \bar{a}\times\bar{c}\ \ \bar{d}\right]=$

1) $\left(\bar{a}.\bar{b}\right)\left[\bar{a}\ \bar{c}\ \bar{d}\right]$ 2) $\left(\bar{a}.\bar{d}\right)\left[\bar{a}\ \bar{b}\ \bar{c}\right]$

3) $\left(\bar{b}.\bar{c}\right)\left[\bar{a}\ \bar{b}\ \bar{d}\right]$ 4) $\left(\bar{a}.\bar{c}\right)\left[\bar{b}\ \bar{c}\ \bar{d}\right]$

14. If $\bar{a},\bar{b},\bar{c}$ are non-zero, non-coplanar vectors then $\left\{\bar{a}\times\left(\bar{b}+\bar{c}\right)\right\}\times\left\{\bar{b}\times\left(\bar{c}-\bar{a}\right)\right\}$ is collinear with the vector

1) $\bar{a}+\bar{b}+\bar{c}$ 2) $\bar{a}-\bar{b}+\bar{c}$

3) $\bar{a}+\bar{b}-\bar{c}$ 4) $\bar{a}-\bar{b}-\bar{c}$

LEVEL-II-(H.W)-KEY

01) 4	02) 3	03) 2	04) 3	05) 3	06) 1
07) 4	08) 1	09) 3	10) 4	11) 4	12) 1
13) 2	14) 4				

LEVEL-II-(H.W)-HINTS

1. $\left\{\left(\bar{a}\times\bar{b}\right)\times\bar{c}\right\}.\bar{c}=0$

2. $\Rightarrow 2\left(\bar{a}\times\bar{b}\right)=2\left(\bar{i}-\bar{j}\right)$

3. $\Rightarrow\left(\bar{a}.\bar{c}\right)\bar{a}-\left(\bar{a}.\bar{a}\right)\bar{c}+\bar{b}=0$

let $\left(\bar{a}.\bar{c}\right)=\theta\Rightarrow\bar{a}.\bar{c}=2\cos\theta$

$\Rightarrow 2\cos\theta\bar{a}-\bar{c}=-\bar{b}\quad\Rightarrow\left(2\cos\theta\,\bar{a}-\bar{c}\right)^2=\left(-\bar{b}\right)^2$

$\Rightarrow 4\cos^2\theta=3\Rightarrow\cos\theta=\frac{\sqrt{3}}{2}\qquad\therefore\theta=\frac{\pi}{6}$

4. $\Rightarrow\left(\bar{a}.\bar{c}\right)\bar{b}-\left(\bar{a}.\bar{b}\right)\bar{c}+\left(\bar{a}.\bar{b}\right)\bar{c}-\left(\bar{b}.\bar{c}\right)\bar{a}=\frac{1}{2}\bar{a}$

$\Rightarrow\left(\bar{a}.\bar{c}\right)\bar{b}-\left(\left(\bar{b}.\bar{c}\right)+\frac{1}{2}\right)\bar{a}=\bar{0}$

$\Rightarrow\bar{a}.\bar{c}=0,\ \bar{b}.\bar{c}=-\frac{1}{2}$

5. $\left(\bar{a}.\bar{c}\right)\bar{b}-\left(\bar{a}.\bar{b}\right)\bar{c}-\frac{\bar{b}}{2}-\frac{\bar{b}}{2}=\bar{0}$

$\Rightarrow\bar{a}.\bar{c}-\frac{1}{2}=0,\ \bar{a}.\bar{b}+\frac{1}{2}=0$

$\Rightarrow\bar{a}.\bar{c}=\frac{1}{2},\ \bar{a}.\bar{b}=-\frac{1}{2}$

$\cos\beta=\frac{1}{2},\ \cos\alpha=-\frac{1}{2}\ \Rightarrow\beta=60^0,\ \alpha=120^0$

6. $\bar{a}.\bar{c}=\bar{b}.\bar{c}=0$

$\left(\bar{a}\times\bar{b}\right)\times\bar{c}=\left(\bar{c}\times\bar{a}\right)\bar{b}=-\left(\bar{c}.\bar{b}\right)\bar{a}=\bar{0}$

7. $\left(\bar{a}.\bar{c}\right)\bar{b}-\left(\bar{b}.\bar{c}\right)\bar{a}=4\bar{b}-5\bar{a}\Rightarrow\bar{a}.\bar{c}=4,\ \bar{b}.\bar{c}=5$

$\bar{a}\times\left(\bar{b}\times\bar{c}\right)=\left(\bar{a}.\bar{c}\right)\bar{b}-\left(\bar{a}.\bar{b}\right)\bar{c}=4\bar{b}-3\bar{c}$

8. $\sum\left[\left(\bar{j}.\bar{a}\right)\bar{i}-\left(\bar{j}.\bar{i}\right)\bar{a}\right]^2$

$=\sum\left[\left(\bar{j}.\bar{a}\right)\bar{i}\right]^2=a_2^2+a_3^2+a_1^2=\bar{a}^2$

9. $\bar{a}\times\left\{\bar{a}\times\left(\bar{a}\times\bar{b}\right)\right\}=\bar{a}\times\left\{\left(\bar{a}.\bar{b}\right)\bar{a}-\left(\bar{a}.\bar{a}\right)\bar{b}\right\}$

$=\left(\bar{a}.\bar{a}\right)\left(\bar{b}\times\bar{a}\right)$

10. $\bar{b}=\bar{c}$

11. $\left(\bar{b}\times\bar{c}\right).\left(\bar{b}\times\bar{c}\right)+\left(\bar{b}.\bar{c}\right)^2=\left|\bar{b}\times\bar{c}\right|+\left(\bar{b}.\bar{c}\right)^2=\left|\bar{b}\right|^2\left|\bar{c}\right|^2$

12. $\left[\bar{a}\ \bar{b}\ \bar{d}\right]\bar{c}-\left[\bar{a}\ \bar{b}\ \bar{c}\right]\bar{d}$

13. $\left[\bar{a}\times\bar{b}\ \ \bar{a}\times\bar{c}\ \ \bar{d}\right]=\left(\bar{a}\times\bar{b}\right).\left\{\left(\bar{a}\times\bar{c}\right)\times\bar{d}\right\}$

$=\left(\bar{a}\times\bar{b}\right).\left\{\left(\bar{a}.\bar{d}\right)\bar{c}-\left(\bar{c}.\bar{d}\right)\bar{a}\right\}$

$=\left(\bar{a}.\bar{d}\right).\left[\bar{a}\ \ \bar{b}\ \ \bar{c}\right]-0$

14. $\{(\bar{a}\times\bar{b})-(\bar{c}\times\bar{a})\}\times\{(\bar{b}\times\bar{a})+(\bar{a}\times\bar{b})\}$
$=(\bar{a}\times\bar{b})\times(\bar{b}\times\bar{a})+(\bar{a}\times\bar{b})\times(\bar{a}\times\bar{b})$
$+(\bar{b}\times\bar{a})\times(\bar{c}\times\bar{a})+(\bar{a}\times\bar{b})\times(\bar{c}\times\bar{a})$

LEVEL - III

1. **If $\bar{a},\bar{b},\bar{c}$ be the three vectors such that $\bar{a}\times\bar{b}=\bar{c}$ and $\bar{b}\times\bar{c}=\bar{a}$ then**

1) $|\bar{a}|=|\bar{b}|=|\bar{c}|=1$ 2) $|\bar{a}|=|\bar{b}|=|\bar{c}|\neq 1$
3) $\bar{a},\bar{b},\bar{c}$ are orthogonal in pairs
4) $|\bar{a}|\neq|\bar{b}|\neq|\bar{c}|$

2. $\bar{a}\times\{\bar{b}\times(\bar{c}\times\bar{a})+(\bar{p}\times\bar{q})\}=$

1) $(\bar{a}.\bar{q})\bar{p}-(\bar{a}.\bar{p})\bar{q}+(\bar{b}.\bar{a})(\bar{a}\times\bar{c})-(\bar{b}\times\bar{c})$
2) $(\bar{a}.\bar{q})\bar{p}-(\bar{a}.\bar{p})\bar{q}+(\bar{b}.\bar{a})(\bar{a}\times\bar{c})$
3) $\bar{a}\times(\bar{p}\times\bar{q})+[\bar{a}\,\bar{b}\,\bar{c}]\bar{c}$ 4) $\bar{a}\times(\bar{p}\times\bar{a})$

3. **The vectors $\bar{a}$ and $\bar{b}$ are mutually perpendicular then** $\bar{a}\times\{\bar{a}\times\{\bar{a}\times(\bar{a}\times\bar{b})\}\}=$

1) $|\bar{a}|^2\bar{b}$ 2) $|\bar{b}|^4\bar{a}$ 3) $|\bar{a}|^4\bar{b}$ 4) $|\bar{a}|\bar{b}$

4. **Let $\bar{a}=\bar{i}-\bar{j},\bar{b}=\bar{j}-\bar{k},\bar{c}=\bar{k}-\bar{i}$. If $\bar{d}$ is a unit vector such that $\bar{a}.\bar{d}=0=[\bar{b}\ \bar{c}\ \bar{d}]$, then $\bar{d}$ is equal to**

1) $\pm\left(\frac{\bar{i}+\bar{j}-2\bar{k}}{\sqrt{6}}\right)$ 2) $\pm\left(\frac{\bar{i}+\bar{j}-\bar{k}}{\sqrt{3}}\right)$
3) $\pm\left(\frac{\bar{i}+\bar{j}+\bar{k}}{\sqrt{3}}\right)$ 4) $\pm\bar{k}$

5. **The pairs $\bar{a},\bar{b}$ and $\bar{c},\bar{d}$ are such that each determines a plane. Then the planes are perpendicular if**

1) $(\bar{a}.\bar{c})(\bar{b}.\bar{d})-(\bar{a}.\bar{d})(\bar{b}.\bar{c})=0$
2) $(\bar{a}.\bar{c})(\bar{a}.\bar{d})-(\bar{b}.\bar{c})(\bar{b}.\bar{d})=0$
3) $(\bar{a}.\bar{c})(\bar{b}.\bar{d})+(\bar{a}.\bar{d})(\bar{b}.\bar{c})=0$
4) $(\bar{a}.\bar{d})(\bar{b}.\bar{c})=(\bar{a}.\bar{b})(\bar{c}.\bar{d})=0$

6. **Given that $\bar{c}$ is in the plane of $\bar{a},\bar{b}$ and $\bar{d}$ is not in the plane of $\bar{a},\bar{b}$ then $(\bar{a}\times\bar{b})\times(\bar{c}\times\bar{d})$ is**

1) in the plane of $\bar{a},\bar{b}$ but not in the plane of $\bar{c},\bar{d}$
2) parallel to $\bar{a}$ 3) parallel to $\bar{c}$ 4) parallel to $\bar{b}$

7. **Unit vectors $\bar{a},\bar{b},\bar{c}$ are coplanar. A unit vector $\bar{d}$ is perpendicular to them. If $(\bar{a}\times\bar{b})\times(\bar{c}\times\bar{d})=\frac{1}{6}\bar{i}-\frac{1}{3}\bar{j}+\frac{1}{3}\bar{k}$ and the angle between $\bar{a}$ and $\bar{b}$ is 30^0 then $\bar{c}=$**

1) $\frac{\bar{i}-2\bar{j}+2\bar{k}}{3}$ 2) $\frac{2\bar{i}+\bar{j}-\bar{k}}{3}$
3) $\frac{-\bar{i}+2\bar{j}+3\bar{k}}{3}$ 4) $\frac{-\bar{i}+2\bar{j}+\bar{k}}{3}$

8. **If $\bar{a}=\bar{i}+\bar{j},\bar{b}=\bar{i}+\bar{k}$, then** $\{\{(\bar{a}\times\bar{b})\times\bar{a}\}\times\bar{a}\}\times\bar{b}=$

1) $\bar{i}+\bar{j}+\bar{k}$ 2) $2(\bar{i}+2\bar{j}-\bar{k})$
3) $-\bar{i}+\bar{j}+\bar{k}$ 4) $\bar{i}-\bar{j}+\bar{k}$

9. **Let $\bar{a}$ be a unit vector and $\bar{b}$ be a non-zero vector not parallel to $\bar{a}$. If two sides of a triangle are represented by the vectors $\sqrt{3}(\bar{a}\times\bar{b})$ and $\bar{b}-(\bar{a}.\bar{b})\bar{a}$ then the angles of the triangle are**

1) $90^0,60^0,30^0$ 2) $45^0,45^0,90^0$
3) $60^0,60^0,60^0$ 4) $75^0,45^0,60^0$

10. **If $\bar{b},\bar{c},\bar{d}$ are non-coplanar vectors then the vector $(\bar{a}\times\bar{b})\times(\bar{c}\times\bar{d})+(\bar{a}\times\bar{c})\times(\bar{d}\times\bar{b})+(\bar{a}\times\bar{d})\times(\bar{b}\times\bar{c})$ is parallel to**

1) $\bar{a}$ 2) $\bar{b}$ 3) $\bar{c}$ 4) $\bar{a}+\bar{b}$

11. **Let $\bar{a}$ be a unit vector and $\bar{b}$ be a non-zero vector not parallel to $\bar{a}$. If $\bar{u}=\sqrt{3}(\bar{a}\times\bar{b})$ and $\bar{v}=(\bar{a}\times\bar{b})\times\bar{a}$ then $(\bar{u},\bar{v})=$**

1) 0^0 2) 30^0 3) 90^0 4) 60^0

12. **In a regular tetrahedron the angle between any two faces is**

1) $\sin^{-1}\frac{1}{3}$ 2) $\cos^{-1}\frac{1}{3}$ 3) $\tan^{-1}\frac{1}{\sqrt{3}}$ 4) $\cos^{-1}\frac{1}{\sqrt{3}}$

13. **Let $\bar{a},\bar{b}$ and $\bar{c}$ be non-zero vectors such that $\left(\bar{a}\times\bar{b}\right)\times\bar{c}=\frac{1}{3}|\bar{b}||\bar{c}|\bar{a}$. If θ is the acute angle between the vectors $\bar{b}$ and $\bar{c}$, then $\sin\theta=$** **(AIE-2004)**

1) $\frac{1}{3}$ 2) $\frac{2\sqrt{2}}{3}$ 3) $\frac{2}{3}$ 4) $\frac{\sqrt{2}}{3}$

14. $\left[\left(\bar{a}\times\bar{b}\right)\times\left(\bar{b}\times\bar{c}\right)\left(\bar{b}\times\bar{c}\right)\times\left(\bar{c}\times\bar{a}\right)\left(\bar{c}\times\bar{a}\right)\times\left(\bar{a}\times\bar{b}\right)\right]=$

1) 0 2) $4\left[\bar{a}\,\bar{b}\,\bar{c}\right]$ 3) $\left[\bar{a}\,\bar{b}\,\bar{c}\right]^4$ 4) $2\left[\bar{a}\,\bar{b}\,\bar{c}\right]$

15. **Let $\bar{a}=\bar{j}-\bar{k},\bar{c}=\bar{i}-\bar{j}-\bar{k}$ then vector $\bar{b}$ satisfying $\bar{a}\times\bar{b}+\bar{c}=\bar{0}$ and $\bar{a}.\bar{b}=3$ is**

1) $-\bar{i}+\bar{j}-2\bar{k}$ 2) $2\bar{i}-\bar{j}+2\bar{k}$

3) $\bar{i}-\bar{j}-2\bar{k}$ 4) $\bar{i}+\bar{j}-2\bar{k}$

LEVEL-III-KEY

01) 3	02) 2	03) 3	04) 1	05) 1	06) 3
07) 1	08) 2	09) 1	10) 1	11) 3	12) 2
13) 2	14) 3	15) 1			

LEVEL-III-HINTS

1. $\bar{a}\times\bar{b}=\bar{c}$ $\therefore$ $\bar{c}$ is $\perp$r $\bar{a}$ and $\bar{b}$, also $\bar{b}\times\bar{c}=\bar{a}$ $\therefore$ $\bar{a}$ is $\perp$r $\bar{b}$ and $\bar{c}$
$\therefore\bar{a},\bar{b},\bar{c}$ are mutually perpendicular

2. $\bar{a}\times\left\{\bar{b}\times\left(\bar{c}\times\bar{a}\right)+\left(\bar{p}\times\bar{q}\right)\right\}$
$\bar{a}\times\left\{\left(\bar{b}.\bar{a}\right)\bar{c}-\left(\bar{b}.\bar{c}\right)\bar{a}+\bar{p}\times\bar{q}\right\}$
$\left(\bar{b}.\bar{a}\right)\left(\bar{a}\times\bar{c}\right)+\left(\bar{a}.\bar{q}\right)\bar{p}-\left(\bar{a}.\bar{p}\right)\bar{q}$

3. $\bar{a}\times\left\{\left(\bar{a}.\bar{b}\right)\bar{a}-\left(\bar{a}.\bar{a}\right)\bar{b}\right\}=|\bar{a}|^2\left(\bar{b}\times\bar{a}\right)$
$\bar{a}\times\left\{\bar{a}\times\left(\bar{a}\times\left(\bar{b}\times\bar{a}\right)\right)\right\}=\bar{a}\times|\bar{a}|^2\left(\bar{b}\times\bar{a}\right)=|\bar{a}|^4\,\bar{b}$

4. $\bar{d}$ is a vector perpendicular to $\bar{a}$ and coplanar with $\bar{b}$ and $\bar{c}$. Hence $\bar{d}$ is a vector collinear with $\bar{a}\times\left(\bar{b}\times\bar{c}\right)$ $\Rightarrow\bar{d}=\pm\frac{\bar{a}\times\left(\bar{b}\times\bar{c}\right)}{\left|\bar{a}\times\left(\bar{b}\times\bar{c}\right)\right|}$

5. $\bar{a}\times\bar{b}$ and $\bar{c}\times\bar{d}$ are perpendicular
$\Rightarrow\left(\bar{a}\times\bar{b}\right).\left(\bar{c}\times\bar{d}\right)=0$

6. Parallel to $\bar{c}$

7. $\bar{a},\bar{b},\bar{c}$ are coplanar $\Rightarrow\left[\bar{a}\;\bar{b}\;\bar{c}\right]=0$.
$\left[\bar{a}\;\bar{b}\;\bar{d}\right]\bar{c}-\left[\bar{a}\;\bar{b}\;\bar{c}\right]\bar{d}=\frac{1}{6}\bar{i}-\frac{1}{3}\bar{j}+\frac{1}{3}\bar{k}$
$\Rightarrow\left[\left(\bar{a}\times\bar{b}\right).\bar{d}\right]\bar{c}=\frac{1}{6}\bar{i}-\frac{1}{3}\bar{j}+\frac{1}{3}\bar{k}$

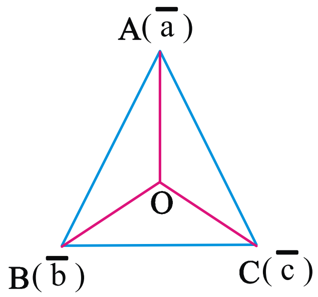
8. $|\bar{a}|^2\left\{\left(\bar{b}.\bar{b}\right)\bar{a}-\left(\bar{a}.\bar{b}\right)\bar{b}\right\}$

9. Two sides are $\sqrt{3}\left(\bar{a}\times\bar{b}\right),\left(\bar{a}\times\bar{b}\right)\times\bar{a}$.
Observe that $\sqrt{3}\left(\bar{a}\times\bar{b}\right).\left\{\left(\bar{a}\times\bar{b}\right)\times\bar{a}\right\}=0$
$\Rightarrow$ angle between these two sides is 90^0.
Lengths of these two sides are in the ratio $\sqrt{3}:1$.
So the remaining angles are $60^0,30^0$.

10. By using vectors product of four vectors formula, the value of the given vector is $-2\left[\bar{b}\;\bar{c}\;\bar{d}\right]\bar{a}$. So it is parallel to $\bar{a}$.

11. $\bar{u}.\bar{v}=\sqrt{3}\left\{\left(\bar{a}\times\bar{b}\right).\left[\left(\bar{a}\times\bar{b}\right)\times\bar{a}\right]\right\}$
$\therefore\bar{u},\bar{v}$ are perpendicular vectors.

12. Let OABC be the regular Tetrahedron and $\bar{a},\bar{b},\bar{c}$ be the position vectors of A, B, C.
OA = OB = OC = AB = BC = CA
$\Rightarrow|\bar{a}|=|\bar{b}|=|\bar{c}|=|\bar{b}-\bar{a}|=|\bar{c}-\bar{b}|=|\bar{a}-\bar{b}|$
Let θ be the angle between the faces OBC and OCA. $\Rightarrow\cos\theta=\left|\frac{\left(\bar{b}\times\bar{c}\right).\left(\bar{c}\times\bar{a}\right)}{|\bar{b}\times\bar{c}||\bar{c}\times\bar{a}|}\right|$ ...... (1)
Observe that $\bar{a}.\bar{b}=\bar{b}.\bar{c}=\bar{c}.\bar{a}$
$=|\bar{a}||\bar{b}|\cos\frac{\pi}{3}=\frac{1}{2}|\bar{a}|^2$
$|\bar{a}\times\bar{b}|=|\bar{b}\times\bar{c}|=|\bar{c}\times\bar{a}|=|\bar{a}||\bar{a}|\sin\frac{\pi}{3}=\frac{\sqrt{3}}{2}|\bar{a}|^2$



13. Use $(\bar{a}\times\bar{b})\times\bar{c}=(\bar{a}.\bar{c})\bar{b}-(\bar{b}.\bar{c})\bar{a}$

14. $[\bar{a}\,\bar{b}\,\bar{c}]^2[\bar{a}\,\bar{b}\,\bar{c}]^2$

15. $\bar{a}\times\bar{b}+\bar{c}=\bar{0} \quad \Rightarrow \bar{a}\times(\bar{a}\times\bar{b})+(\bar{a}\times\bar{c})=\bar{0}$

$(\bar{a}.\bar{b})\bar{a}-(\bar{a}.\bar{a})\bar{b}+(\bar{a}\times\bar{c})=\bar{0}$

$3\bar{a}-2\bar{b}+\bar{a}\times\bar{c}=\bar{0}$

$\Rightarrow 2\bar{b}=3\bar{a}+\bar{a}\times\bar{c}=\bar{0}$

$\therefore \bar{b}=-\bar{i}+\bar{j}-2\bar{k}$

LEVEL - IV

1. Match the following

I. $(\bar{a}\times\bar{b})\times\bar{c}+(\bar{b}\times\bar{c})\times\bar{a}+(\bar{c}\times\bar{a})\times\bar{b}$ — a) $\bar{0}$

II. $[\bar{a}+2\bar{b}\;\; 2\bar{b}+\bar{c}\;\; \bar{a}+5\bar{c}]$ — b) $(\bar{a}.\bar{d})[\bar{a}\;\bar{b}\;\bar{c}]$

III. $(\bar{a}\times\bar{b})\times(\bar{b}\times\bar{c})$ — c) $[\bar{a}\;\bar{b}\;\bar{c}]\bar{b}$

IV. $[(\bar{a}\times\bar{b})\times(\bar{a}\times\bar{c})].\bar{d}$ — d) $12[\bar{a}\;\bar{b}\;\bar{c}]$

1) c,a,b,d 2) a,d,c,b
3) b,c,d,a 4) b,a,c,d

2. Match the following

List-I	List-II
A) $[(\bar{a}\times\bar{b})\times(\bar{a}\times\bar{c})].\bar{d}$	1) $(\bar{a}.\bar{a})(\bar{b}\times\bar{a})$
B) $[\bar{b}\times\bar{c}\;\; \bar{c}\times\bar{a}\;\; \bar{a}\times\bar{b}]$	2) $[\bar{a}\;\bar{b}\;\bar{c}]^4$
C) $\bar{a}\times[\bar{a}\times(\bar{a}\times\bar{b})]$	3) $[\bar{a}\;\bar{b}\;\bar{c}](\bar{a}.\bar{d})$
D) If $\bar{a}\parallel\bar{b}\times\bar{c}$ then	4) $[\bar{a}\;\bar{b}\;\bar{c}]^2$
$(\bar{a}\times\bar{b}).(\bar{a}\times\bar{c})=$	5) $\lvert\bar{a}\rvert^2(\bar{b}.\bar{c})$

1) A -2, B - 4, C - 1, D - 3
2) A - 3, B - 2, C - 5, D - 4
3) A - 3, B - 4, C - 1, D - 5
4) A - 5, B - 4, C - 1, D - 3

3. Statement-I : $\bar{a}\times(\bar{b}\times\bar{c}),\bar{b}\times(\bar{c}\times\bar{a}),\bar{c}\times(\bar{a}\times\bar{b})$ **are coplanar vectors.**

Statement-II : If there exists scalars ℓ, m, n not all zero's such that $\ell\bar{a}+m\bar{b}+n\bar{c}=\bar{0}$ then the vectors $\bar{a},\bar{b},\bar{c}$ are coplanar.

1) Both I and II are true and II is the correct explanation of I
2) Both I and II are true but II is not correct explanation of I.
3) I is true, II is false
4) I is false, II is true

4. Statement-I : $\bar{a},\bar{b},\bar{c}$ are non-coplanar vectors. Then

$[(\bar{a}\times\bar{b})\times(\bar{b}\times\bar{c})\;\;(\bar{b}\times\bar{c})\times(\bar{c}\times\bar{a})\;\;(\bar{c}\times\bar{a})\times(\bar{a}\times\bar{b})]$

$=[\bar{a}\;\bar{b}\;\bar{c}]$

Statement-II : If $\bar{a},\bar{b},\bar{c}$ are three non coplanar vectors then

$[[\bar{a}\;\bar{b}\;\bar{c}](\bar{a}\times\bar{b})\;\;[\bar{b}\;\bar{c}\;\bar{a}](\bar{b}\times\bar{c})\;\;[\bar{c}\;\bar{a}\;\bar{b}](\bar{c}\times\bar{a})]$

$=[\bar{a}\;\bar{b}\;\bar{c}]^3$

1) Only I is true 2) Only II is true
3) Both I & II are true
4) Neither I nor II are true

LEVEL-IV-KEY

01) 2 02) 3 03) 1 04) 4

LEVEL-IV-HINTS

1. i) Use definition of vector product of three vectors.
ii) Use definition of scalar triple product.
iii) Use definition of vector product of four vectors
iv) Use definition of vector product of four vectors

2. 1) $[(\bar{a}\times\bar{b})\times(\bar{a}\times\bar{c})].\bar{d}=[[\bar{a}\,\bar{b}\,\bar{c}]\bar{a}-\bar{0}].\bar{d}$

2) $[\bar{b}\times\bar{c}\;\; \bar{c}\times\bar{a}\;\; \bar{a}\times\bar{b}]=[\bar{a}\;\bar{b}\;\bar{c}]^2$

3) $\bar{a}[\bar{a}\times(\bar{a}\times\bar{b})]=\bar{a}[(\bar{a}.\bar{b})\bar{a}-\lvert\bar{a}\rvert^2\bar{b}]$

$=\lvert\bar{a}\rvert^2(\bar{b}\times\bar{a})$

3. Let $\bar{p}=\bar{a}\times(\bar{b}\times\bar{c})$, $\bar{q}=\bar{b}\times(\bar{c}\times\bar{a})$,

$\bar{r}=\bar{c}\times(\bar{a}\times\bar{b})$

$\bar{p}+\bar{q}+\bar{r}=\bar{0}$.

4. $[\bar{a}\times\bar{b}\;\; \bar{b}\times\bar{c}\;\; \bar{c}\times\bar{a}]=[\bar{a}\;\bar{b}\;\bar{c}]^2$

and $[l\bar{a}\;\; m\bar{b}\;\; n\bar{c}]=lmn[\bar{a}\,\bar{b}\,\bar{c}]$